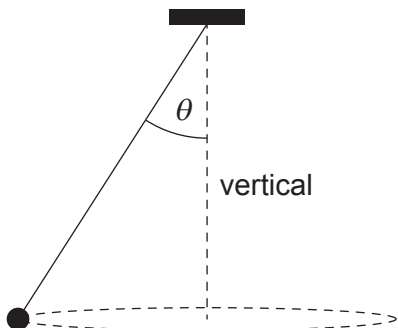


PAPER 1 / QUESTION 89

2 May 2023 / TZ1 (file label) / HL Paper 1 / original question 6 / 2223-6507

89. A ball attached to a string is made to rotate with constant speed along a horizontal circle. The string is attached to the ceiling and makes an angle of θ° with the vertical. The tension in the string is T .



What is correct about the horizontal component and vertical component of the net force on the ball?

	Horizontal component	Vertical component
A.	$T \cos \theta$	$T \sin \theta$
B.	$T \sin \theta$	$T \cos \theta$
C.	$T \cos \theta$	0
D.	$T \sin \theta$	0